

# Лабораторная Работа №6. Статическая маршрутизация VLAN

Администрирование локальных сетей

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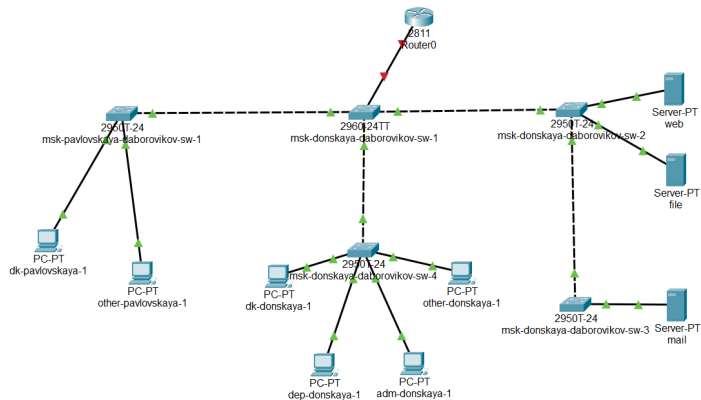
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Настроить статическую маршрутизацию VLAN в сети.

# Размещение маршрутизатора



**Figure 1:** Размещение маршрутизатора Cisco 2811

# Конфигурирование маршрутизатора

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-daborovikov-gw-1
msk-donskaya-daborovikov-gw-1(config)#line vty 0 4
msk-donskaya-daborovikov-gw-1(config-line)#password cisco
msk-donskaya-daborovikov-gw-1(config-line)#login
msk-donskaya-daborovikov-gw-1(config-line)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#line console 0
msk-donskaya-daborovikov-gw-1(config-line)#password cisco
msk-donskaya-daborovikov-gw-1(config-line)#login
msk-donskaya-daborovikov-gw-1(config-line)#enable secret cisco
msk-donskaya-daborovikov-gw-1(config)#service password-encryption
msk-donskaya-daborovikov-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-daborovikov-gw-1(config)#ip domain name donskeya.rudn.edu
msk-donskaya-daborovikov-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-daborovikov-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-daborovikov-gw-1(config)#line vty 0 4
*Mar 1 0:4:13.100: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-daborovikov-gw-1(config-line)#transport input ssh
msk-donskaya-daborovikov-gw-1(config-line)#
```

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**Figure 2:** Конфигурирование маршрутизатора, удаленной подключение ssh

# trunk-порт 24

```

Password:

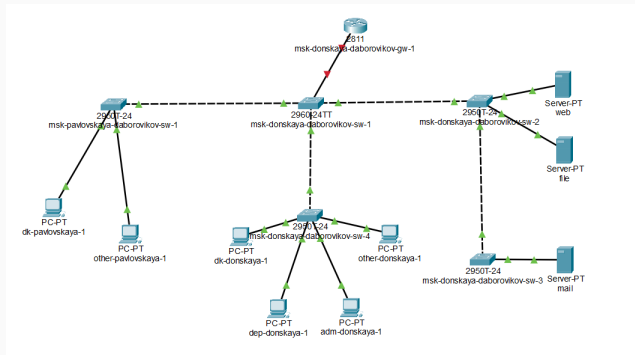
msk-donskaya-daborovikov-sw-1>enable
Password:
Password:
msk-donskaya-daborovikov-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-1(config)#interface f0/24
msk-donskaya-daborovikov-sw-1(config-if)#switchport mode trunk
msk-donskaya-daborovikov-sw-1(config-if)#^Z
msk-donskaya-daborovikov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-daborovikov-sw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-1#
```

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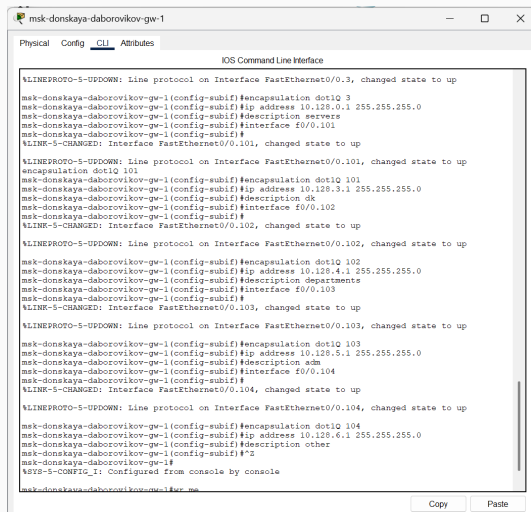
**Figure 3:** Порт 24 как trunk-порт

# Переименование маршрутизатора



**Figure 4:** Переименование маршрутизатора

# Настройка виртуальных интерфейсов



```
msk-donskaya-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
msk-donskaya-daborovikov-gw-1(config-subif)#encapsulation dot1Q 3
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-daborovikov-gw-1(config-subif)#description servers
msk-donskaya-daborovikov-gw-1(config-subif)#interface f0/0.101
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.101, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up
encapsulation dot1Q 101
msk-donskaya-daborovikov-gw-1(config-subif)#encapsulation dot1Q 101
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-daborovikov-gw-1(config-subif)#description dk
msk-donskaya-daborovikov-gw-1(config-subif)#interface f0/0.102
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.102, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

msk-donskaya-daborovikov-gw-1(config-subif)#encapsulation dot1Q 102
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
msk-donskaya-daborovikov-gw-1(config-subif)#description departments
msk-donskaya-daborovikov-gw-1(config-subif)#interface f0/0.103
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.103, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up

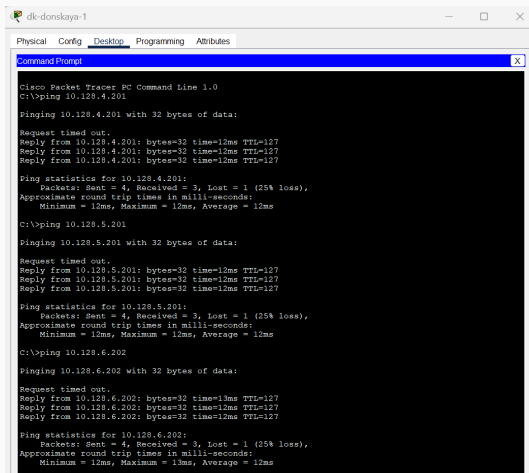
msk-donskaya-daborovikov-gw-1(config-subif)#encapsulation dot1Q 103
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
msk-donskaya-daborovikov-gw-1(config-subif)#description adm
msk-donskaya-daborovikov-gw-1(config-subif)#interface f0/0.104
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.104, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

msk-donskaya-daborovikov-gw-1(config-subif)#encapsulation dot1Q 104
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
msk-donskaya-daborovikov-gw-1(config-subif)#description other
msk-donskaya-daborovikov-gw-1(config-subif)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-daborovikov-gw-1#
```

Figure 5: Настроим виртуальные интерфейсы, соответствующие номерам VLAN





```
dk-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.4.201

Pinging 10.128.4.201 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.201: bytes=32 time=12ms TTL=127
Reply from 10.128.4.201: bytes=32 time=12ms TTL=127
Reply from 10.128.4.201: bytes=32 time=12ms TTL=127

Ping statistics for 10.128.4.201:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 12ms, Average = 12ms

C:\>ping 10.128.5.201

Pinging 10.128.5.201 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.201: bytes=32 time=12ms TTL=127
Reply from 10.128.5.201: bytes=32 time=12ms TTL=127
Reply from 10.128.5.201: bytes=32 time=12ms TTL=127

Ping statistics for 10.128.5.201:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 12ms, Average = 12ms

C:\>ping 10.128.6.202

Pinging 10.128.6.202 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.202: bytes=32 time=13ms TTL=127
Reply from 10.128.6.202: bytes=32 time=12ms TTL=127
Reply from 10.128.6.202: bytes=32 time=12ms TTL=127

Ping statistics for 10.128.6.202:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 13ms, Average = 12ms
```

Figure 6: ping

# Процесс передвижения пакета ICMP по сети.

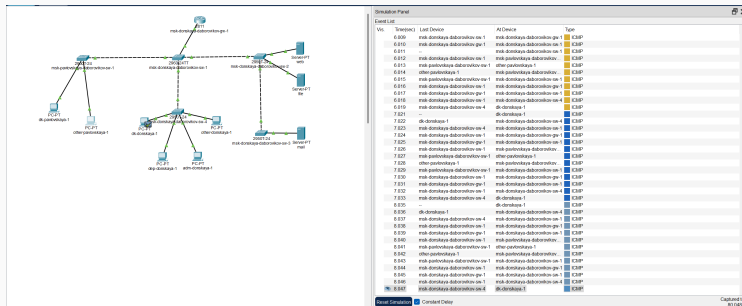


Figure 7: Процесс передвижения пакета ICMP по сети.

# Содержимое передаваемого пакета

PDU Information at Device: msk-donskaya-daborovikov-gw-1

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: msk-donskaya-daborovikov-gw-1  
Source: dk-donskaya-1  
Destination: 10.128.6.202

In Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 10.128.6.202, Dest. IP: 10.128.3.201  
ICMP Message Type: 0

Layer 2: Dot1q Header  
0001.C727.0150 >> 0060.3E56.8501

Layer 1: Port FastEthernet0/0

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 10.128.6.202, Dest. IP: 10.128.3.201  
ICMP Message Type: 0

Layer 2: Dot1q Header  
0060.3E56.8501 >> 0060.5C43.ABA6

Layer 1: Port(s): FastEthernet0/0

1. FastEthernet0/0 receives the frame.

Figure 8: Содержимое передаваемого пакета

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В ходе выполнения лабораторной работы я приобрел практические навыки по настройке статической маршрутизации VLAN в сети.